

Cross-Rover Semantic Segmentation of Mars NavCam Terrain with Domain Adaptation

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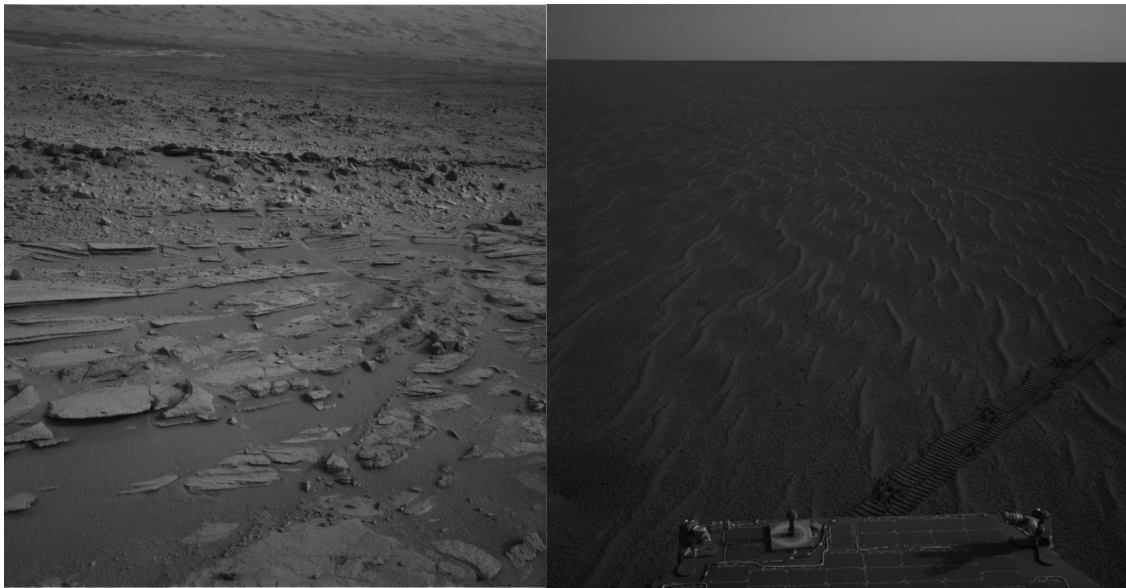
AI4MARS · MSL→MER · ResNet34-U-Net · July 2026

Abstract. This work investigates semantic segmentation of Mars navigation images into four terrain classes (*soil*, *bedrock*, *sand*, *big_rock*) using the AI4MARS dataset. The proposed project pretrains a ResNet34 U-Net on MSL (Curiosity images) and adapts it to MER (Spirit/Opportunity images) through unsupervised domain adaptation (UDA) or a semi-supervised approach. The infrequent *big_rock* class stays the most challenging, despite strong transfer performance on the more common classes.

1 Motivation and Rationale

Autonomous Mars rovers like Curiosity (MSL) and Spirit/Opportunity (MER) use navigation cameras for detection, path planning, and terrain segmentation. Terrain segmentation maps each image to terrain types, allowing the rover to trace the optimal path.

The AI4MARS dataset [1] offers expert labels from multiple Mars missions. However, a domain shift exists between missions due to differences in camera geometry, lighting, and terrain types across rovers' cameras. Models trained only on MSL data underperform on MER and do not reach the same performance level as on MSL.



(a) MSL (Curiosity, rock formations)

(b) MER (sand-rich, darker terrain)

Figure 1: Cross-mission appearance shift in NavCam imagery (AI4MARS).

1.1 Why this matters

Manual annotation of Mars images for segmentation is resource intensive, and MER has substantially fewer labeled NavCam frames than MSL. We employ unsupervised domain adaptation (UDA) and semi-supervised learning to leverage unlabeled MER images together with a smaller labeled MER set for semi-supervised training. Pretraining uses the more abundant MSL data. Due to computational constraints, we use only a subset of the AI4MARS dataset.

1.2 Rationale for pipeline design

Instead of training a U-Net from scratch, a ResNet34 encoder is utilized. By implementing this choice we can fine-tune on the specific dataset and avoid the computational burden of training on big datasets, since ResNet34 is already pretrained on ImageNet.

1.3 Problem

In this work, we try to solve the challenge of adapting models from one Mars rover mission (MSL, with labeled data) to another (MER, where most data is unlabeled) using unsupervised domain adaptation (UDA). Both domains share a common set of four terrain classes and one additional null class. Our goal is to train a model that performs well on the target domain (MER) without ever seeing the MER test labels. At the same time, we aim to maintain strong performance on the source domain (MSL).

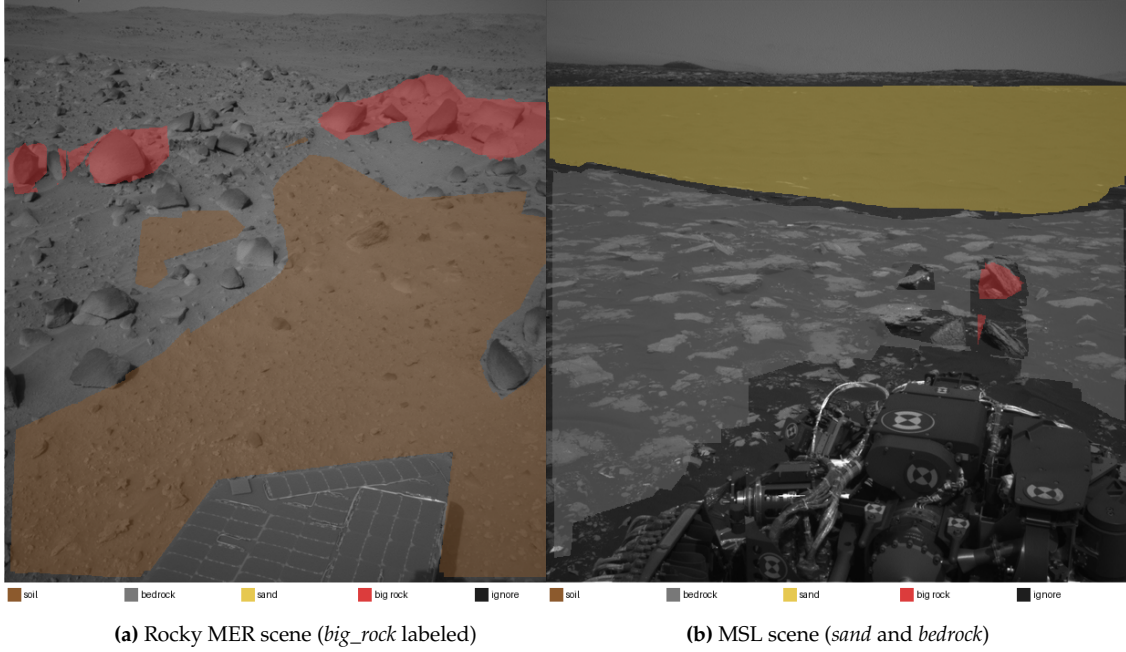


Figure 2: MER training examples with colored class overlays (image + mask).

2 State of the Art

2.1 ViT models and foundation models

2.1.1 Vision Transformers (ViT)

Vision Transformers (ViT) [4] approach image understanding by dividing each image into fixed-size patches, which are processed as a sequence by transformer encoder blocks. Hybrid models like SegFormer capture long-range dependencies while integrating multi-scale features. However ViT architectures require extensive pretraining and are more resource-demanding than CNNs.

2.1.2 Segment Anything Model (SAM)

SAM [12] is a promptable segmentation foundation model developed by Meta and trained on over 1 billion masks. Given a point, box, or prompt, it produces high-quality object masks with strong zero-shot generalization on natural images. Follow-ups (SAM 2, domain-specific fine-tunes) extend this idea to video and specialized sensors.

2.1.3 Relevance to Mars terrain segmentation

While SAM and ViT-based pipelines represent SOTA segmentation architectures, our project faces important differences and constraints. AI4MARS defines four terrain classes, not prompt-based instance masks as in

SAM. Adaptation from labeled MSL to mostly unlabeled MER is the main difficulty. Training budget and labeled MER data are limited, so a ResNet34-U-Net is more practical than a large transformer or SAM. CORAL, MMD, and DANN fit CNN encoders by aligning the U-Net bottleneck (`enc4`). Using the same UDA stack on SAM or ViT would need extra work. Foundation models could still serve as frozen encoders or pseudo-label teachers on unlabeled MER frames.

2.2 Unsupervised domain adaptation SOTA

2.2.1 Vision-language models and CLIP

CLIP [6] pretrains joint image and text embeddings, so new categories can be queried at inference with text prompts (class names) instead of extra labeled data. Regarding segmentation and UDA, these models (CLIP, VLMs, etc.) can provide targets for mask generation, pseudo-labels or text-based alignment when both domains share the same class names.

2.2.2 Foundation-model and prompt-based UDA trends

A prevalent trend in modern domain adaptation (UDA) is to use large pretrained models such as CLIP or ViT as fixed backbones, with only minor refinements or fine-tuning of additional layers or tokens for the target domain. When there is scarce labeled data, these models are often used to generate pseudo-labels to facilitate training. There is also increasing interest in employing models like diffusion or SAM to produce coarse masks on new data, which can then be used to train smaller segmentation models alongside a limited set of manually labeled examples.

2.3 Our choices for UDA

We adapt MSL→MER with feature-level methods on shared `enc4` representations. We do **not** use AdaptSegNet, CLIP, or other VLM pipelines in the reported experiments.

- **CORAL** [8]: covariance alignment on dense `enc4` feature maps.
- **MMD** [3]: RBF kernel Minimum Mean Discrepancy on dense `enc4` features.
- **DANN** [2]: domain adversarial discriminator that uses gradient reversal with MSL segmentation loss and unlabeled MER batches.
- **ADDA** [9]: two-phase adversarial adaptation: (1) train a domain discriminator on frozen global average pooled features; (2) adapt a joint encoder with MSL segmentation, MER adversarial alignment, and dense CORAL on `enc4` (phase 2).

CORAL, MMD, and DANN combine segmentation loss with a domain adaptation term ($\lambda=0.1$). ADDA instead uses a dedicated alternate schedule (5 discriminator epochs plus 10 adaptation epochs) with adversarial weight ramped by the Ganin schedule and CORAL weight 0.1. All checkpoints are selected on `mer_val`.

2.3.1 DANN vs ADDA

DANN updates the encoder and discriminator together in one step by using the gradient reversal. ADDA pretrains the discriminator and then alternates between discriminator and encoder updates. The difference is that ADDA does not use a shared encoder.

2.4 Semi-supervised and pseudo-label segmentation

Self-training assigns pseudo-labels to unlabeled pixels that surpass a confidence threshold. We use EMA teachers [10] to stabilize pseudo-labels. We use joint losses that combine labeled and pseudo terms in the same optimization step to try to outperform other approaches.

3 Objectives

3.1 Main objective

Train a segmenter on MSL and adapt it to MER so that MER test mIoU surpasses that of a zero-shot MSL-pretrained baseline while maintaining reasonable MSL test performance.

3.2 Secondary objectives

1. Compare feature-level UDA and try pseudo-labeling and semi-supervised variants under a common encoder-decoder architecture.
2. Analyze class behavior, especially the rare *big_rock* class.

4 Methodology

4.1 Dataset and preprocessing

We use the AI4MARS dataset [1]: RGB images with four terrain classes (*soil*, *bedrock*, *sand*, *big_rock*). Rover body and uncertain pixels (distant objects, sky, background) are excluded from loss and metrics. Images are resized to 512×512 and ImageNet-normalized. On MSL only, auxiliary rover-body and range masks set additional pixels to ignore, avoiding training on the rover or distant regions where manual annotation was deliberately omitted in AI4MARS. Separate splits are used for MSL pretraining and testing, labeled MER adaptation, unlabeled MER UDA, and MER validation and test evaluation.

Table 1: Data splits and roles.

Split	Role	<i>N</i>
msl_train / val / test	Pretrain / test	3000 / 322
mer_labeled_adapt	Semi-supervised anchor	200
mer_uda	Unlabeled MER (UDA + pseudo)	2000
mer_val / mer_test	Select adaptation / final test	200 / 204

4.1.1 Validation protocol

1. Tune on the appropriate validation split.
2. Early-stop on validation mIoU to avoid overfitting.
3. Report test metrics *once* for selected finalists only.

Pretrain hyperparameter search uses `msl_val`.

Table 2: Pretraining hyperparameter sets (winner: low learning rate).

Set	Encoder LR	Decoder LR	Weight decay
Base	10^{-5}	10^{-4}	10^{-4}
Low LR (selected)	5×10^{-6}	5×10^{-5}	10^{-4}
Higher decoder LR	10^{-5}	2×10^{-4}	10^{-4}

4.2 Metrics

We report mean IoU, per-class IoU, and frequency-weighted IoU (wIoU). The problem with unweighted mIoU is that it can mask saturation on frequent classes. Zero-shot pretraining shows a substantial drop from the MSL test to the MER test, illustrating that the cross-mission domain gap is notable.

4.3 Architecture

We couple an ImageNet-pretrained ResNet34 encoder [5] with a U-Net decoder [7] and a segmentation head. Skip connections link encoder stages to matching decoder levels; then, the bottleneck `enc4` contains the deepest features. We split the forward path (encode + decode) so that feature UDA can target `enc4` without breaking segmentation, allowing feature-level UDA to be applied at this bottleneck.

4.3.1 Encoder-decoder data flow

The input RGB image passes through the ResNet stem and encoder stages, producing skip tensors. The decoder upsamples from `enc4`, concatenating skips at each level and applying convolution blocks at each stage, then a final upsample restores full resolution before the classification head. We apply domain alignment

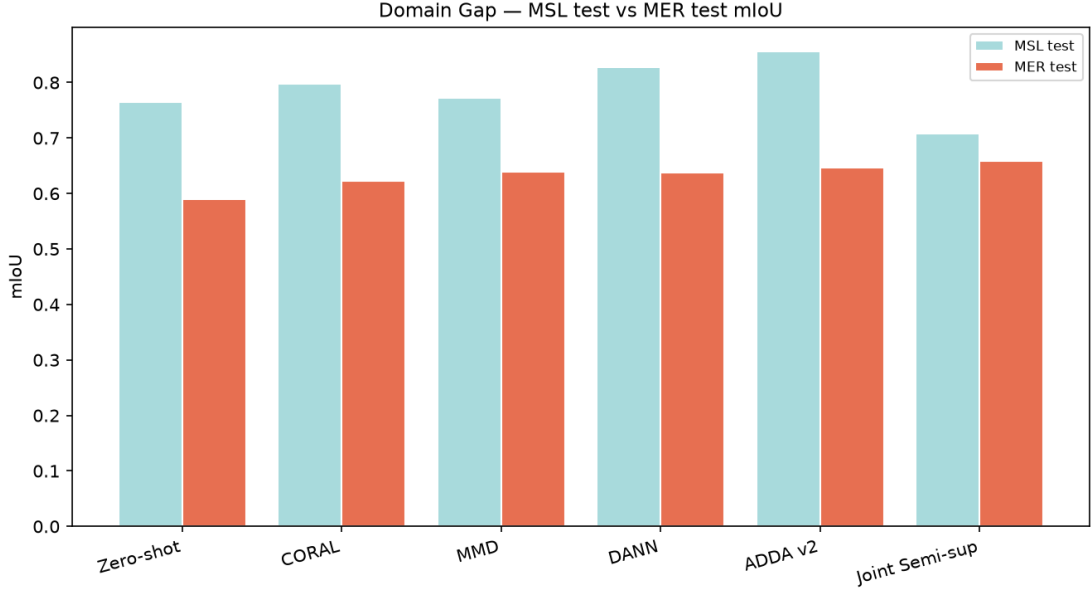


Figure 3: Domain gap: MSL test vs. MER test mIoU (zero-shot pretrain drops ~ 0.17 on MER).

only to enc4. The skip tensors retain source fine structure, helping MSL accuracy but limiting how much MER appearance can change in shallow layers.

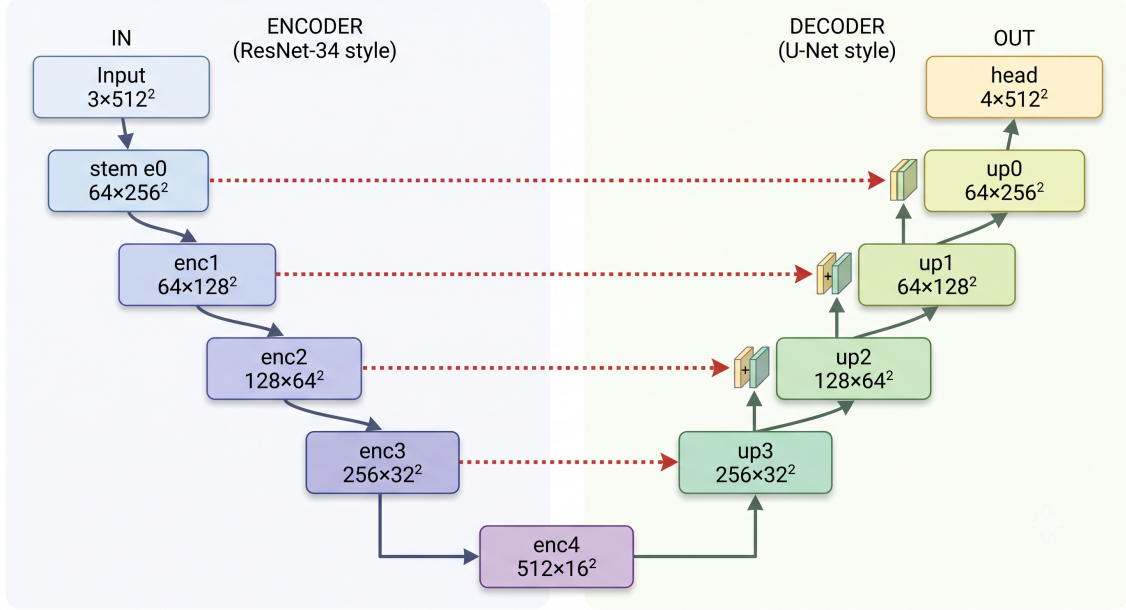


Figure 4: ResNet34-U-Net architecture: ImageNet-pretrained encoder, skip connections at four scales, bottleneck enc4 ($512 \times 16 \times 16$) for feature-level UDA and decoder head producing 512×512 class logits.

4.4 Loss functions and training stages

All training stages minimize cross-entropy on valid pixels. Segmentation loss is the mean cross-entropy over labeled pixels in each batch. Let f_θ denote the segmenter, $\Omega(Y)$ the set of labeled pixels in mask Y , and $\text{CE}(\hat{y}, y)$ the per-pixel cross-entropy. For a batch of images X and labels Y ,

$$\mathcal{L}_{\text{seg}}(X, Y) = \frac{1}{|\Omega(Y)|} \sum_{p \in \Omega(Y)} \text{CE}(f_\theta(X)_p, y_p). \quad (1)$$

Feature UDA methods extract bottleneck enc4 tensors, flatten them to per-location features with normalization and apply domain losses on those features or on global-average-pooled vectors $h(\mathbf{z}) = \text{GAP}(\mathbf{z})$.

4.4.1 Segmentation and pretraining

MSL pretraining minimizes $\mathcal{L}_{\text{seg}}$ on the MSL training split. The best checkpoint on MSL validation initializes all adaptation runs, and then hyperparameter search compares learning rates and weight decay. The selected low learning-rate configuration is used to initialize follow-up semi-supervised runs.

4.4.2 Feature-level UDA

From the pretraining checkpoint, each iteration pairs labeled MSL batches (X_S, Y_S) with unlabeled MER images X_T and optimizes

$$\mathcal{L}_{\text{UDA}} = \mathcal{L}_{\text{seg}}(X_S, Y_S) + \lambda \mathcal{L}_{\text{DA}}, \quad \lambda = 0.1, \quad (2)$$

with $\mathcal{L}_{\text{DA}}$ chosen as follows (all on `enc4 feat_map` tensors unless noted).

CORAL [8] aligns second-order statistics of source and target features. With centered covariance matrices C_S, C_T of $\phi(\mathbf{z}_S)$ and $\phi(\mathbf{z}_T)$,

$$\mathcal{L}_{\text{CORAL}} = \frac{1}{4C} \|C_S - C_T\|_F^2. \quad (3)$$

MMD [3] uses an unbiased RBF-kernel estimate ($\sigma=1.0$, subsample cap 2048):

$$\mathcal{L}_{\text{MMD}} = \mathbb{E}[k(\phi_s, \phi'_s)] + \mathbb{E}[k(\phi_t, \phi'_t)] - 2\mathbb{E}[k(\phi_s, \phi_t)], \quad k(u, v) = \exp\left(-\frac{\|u-v\|^2}{2\sigma^2}\right). \quad (4)$$

DANN [2] attaches a domain discriminator D to `enc4` features with gradient reversal (GRL) strength augmented by the Ganin schedule ($\lambda_{\text{max}}=0.1$):

$$\mathcal{L}_{\text{DANN}} = \mathcal{L}_{\text{seg}}(X_S, Y_S) + \text{BCE}(D(\text{GRL}_{\lambda_p}(h_S)), \mathbf{0}) + \text{BCE}(D(\text{GRL}_{\lambda_p}(h_T)), \mathbf{1}), \quad (5)$$

where source labels are 0 and target labels are 1. The discriminator and encoder are updated jointly at each step.

Checkpoints for CORAL, MMD, and DANN are selected on MER validation.

4.4.3 ADDA adaptation

ADDA adaptation [9] runs in two phases from the same pretrained checkpoint.

Phase 1 (discriminator pre-training). The segmenter is frozen, and the discriminator is trained on `GAP(enc4)` features from paired MSL/MER batches:

$$\mathcal{L}_D = \text{BCE}(D(h_S), \mathbf{0}) + \text{BCE}(D(h_T), \mathbf{1}). \quad (6)$$

Phase 2 (encoder adaptation). Each iteration alternates between a discriminator update on detached features and an encoder/decoder step that combines MSL segmentation loss, adversarial loss, and dense CORAL on `enc4`.

$$\mathcal{L}_{\text{ADDA}} = \lambda_{\text{seg}} \mathcal{L}_{\text{seg}}(X_S, Y_S) + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{CORAL}} \mathcal{L}_{\text{CORAL}}, \quad (7)$$

where $\mathcal{L}_{\text{adv}} = \text{BCE}(D(h_T), \mathbf{0})$ pushes MER features toward the source domain, $\mathcal{L}_{\text{CORAL}}$ is Eq. 3 on dense `enc4` maps, $\lambda_{\text{seg}}=1$, λ_{adv} is ramped (Ganin schedule, peak 0.1) and $\lambda_{\text{CORAL}}=0.1$. The selected recipe uses 5 discriminator epochs, 10 adaptation epochs and batch size 32.

4.4.4 Pseudo-labeling

On unlabeled MER images X_T , the model (or EMA teacher) produces per-pixel class predictions $\hat{y}_p = \arg \max_c f_{\theta}(X_T)_{p,c}$ and confidence $c_p = \max_c \text{softmax}(f_{\theta}(X_T))_{p,c}$. Pixels with $c_p < \tau$ are ignored; batches with too few confident pixels are skipped. Self-training minimizes

$$\mathcal{L}_{\text{pseudo}} = \mathcal{L}_{\text{seg}}(X_T, \hat{Y}_T), \quad (8)$$

on pseudo-labels. When an EMA teacher θ' is used, teacher weights are updated after each step with exponential moving average decay.

4.4.5 Two-phase semi-supervision

The variants first run labeled MER epochs, then pseudo-label epochs on unlabeled MER data; the two objectives are *not* combined in the same backward pass.

4.4.6 Joint semi-supervision

The joint variant combines labeled and pseudo terms in a single optimization step:

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{seg}}(X_T^{\text{lab}}, Y_T^{\text{lab}}) + \lambda_u \mathcal{L}_{\text{pseudo}}(X_T^{\text{uda}}), \quad (9)$$

with an optional EMA teacher, a labeled-only warm-up epoch, and the same confidence filtering as Eq. 8.

5 Experiments and Results

5.1 Experimental setup

All runs share a fixed batch size and AdamW optimizer. Full training used a cloud-hosted GPU for speed.

5.2 Implementation overview

All experiments share a single PyTorch pipeline built around the same ResNet34-U-Net segmenter, and every method reuses it as a baseline, differing only in the training objective (MSL pretraining, feature-level alignment on enc4, pseudo-label self-training, or joint semi-supervision). Checkpoints store only model weights; the best epoch is chosen, and metrics are logged separately for analysis.

5.3 Pretraining

MSL pretraining on the training split reaches strong validation mIoU; the selected checkpoint is evaluated zero-shot on MER before any adaptation (Table 3, zero-shot row).

5.4 Main pipeline

Table 3 summarizes test mIoU and wIoU after one validation-selected evaluation per method. On the main pipeline, zero-shot pretraining establishes a baseline on the MER test. Feature-level UDA methods (CORAL, MMD, DANN, ADDA) improve over zero-shot, with ADDA performing best among pure UDA approaches on weighted IoU. Pure pseudo-labeling collapses without labeled MER anchoring. Two-phase semi-supervision underperforms zero-shot. Joint semi-supervision reaches the best overall MER test mIoU by combining labeled and unlabeled MER data in one optimization step. MMD and DANN also show higher MSL test than zero-shot because UDA continues MSL supervised training during adaptation while selecting on MER validation.

Table 3: Main pipeline: test mIoU and weighted IoU (wIoU).

Method	MSL test			
	mIoU	wIoU	mIoU	wIoU
Zero-shot (pretrain)	0.764	0.958	0.590	0.769
CORAL	0.797	0.955	0.623	0.810
MMD	0.772	0.947	0.639	0.821
DANN	0.828	0.957	0.637	0.820
ADDA	0.857	0.944	0.647	0.846
Pseudo-label	0.136	0.273	0.127	0.203
Semi-sup	0.633	0.688	0.393	0.535

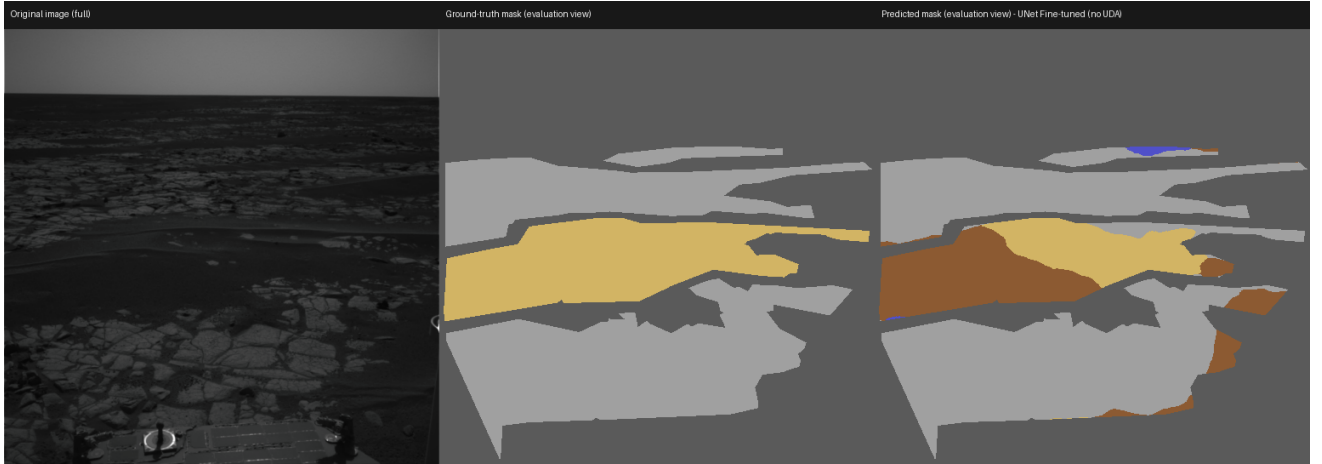


Figure 5: Zero-shot U-Net on MER test (MSL-pretrained, no adaptation): original NavCam frame, ground-truth evaluation mask and predicted mask.

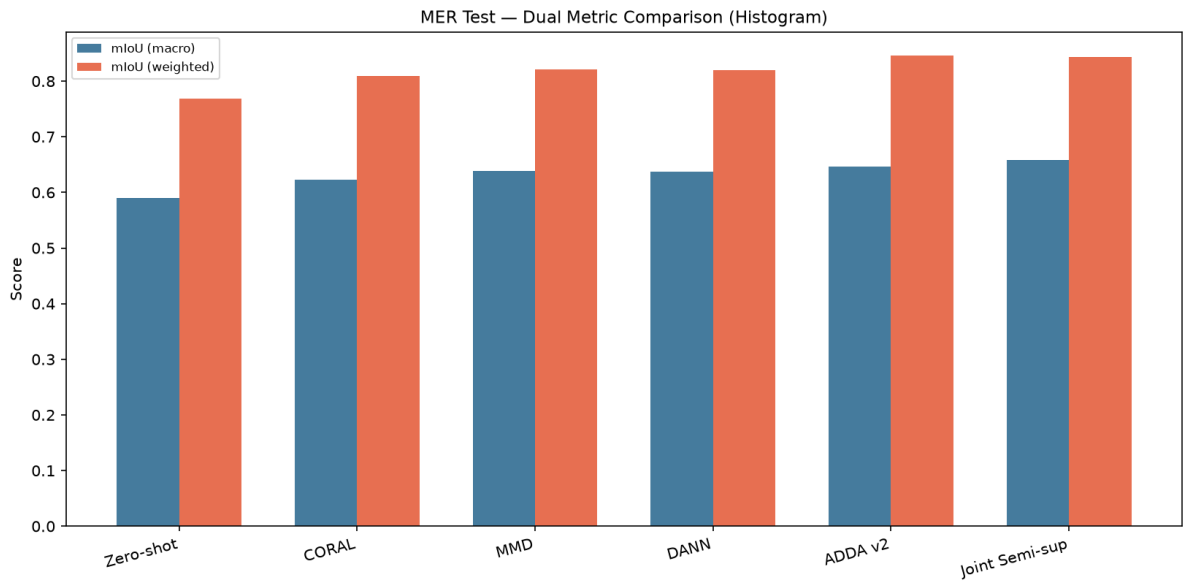


Figure 6: MER test mIoU and wIoU across main-pipeline methods (six key methods).

5.5 Semi-supervised and pseudo-labeling

From the pretraining checkpoint, we tried semi-supervised, pseudo-labeling, and a combination of the two: pseudo-labeling with an EMA teacher, two-phase semi-supervision (labeled epochs then pseudo epochs), and joint semi-supervision. Pseudo-labeling employs an EMA teacher. The student model is updated using the teacher’s labels; joint semi-supervision enhances pseudo-label quality by anchoring training on labeled MER images while simultaneously utilizing unlabeled MER data within the same optimization step. Joint semi-supervision obtains the highest MER test mIoU among all methods tried. Per-class gains are largest for the rare *big_rock* class relative to zero-shot pretraining, though absolute *big_rock* IoU remains low. Table 4 lists follow-up runs; Table 6 reports per-class MER test IoU for joint semi-supervision vs. zero-shot pretrain.

Table 4: Semi-supervised follow-up: MER test mIoU and wIoU.

Method	mIoU	wIoU
Pseudo-label + EMA	0.618	0.803
Pseudo-label + EMA (strict)	0.590	0.772
Semi-sup (two-phase)	0.612	0.810
Joint semi-supervision	0.658	0.844

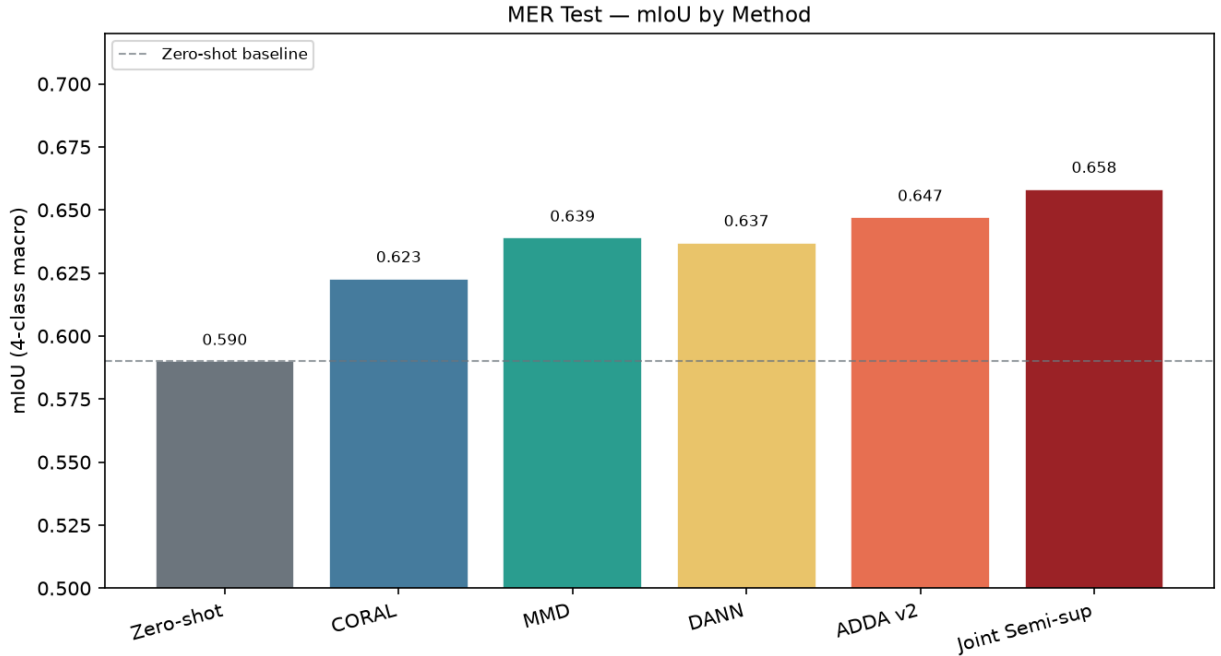


Figure 7: MER test mIoU ranking: zero-shot pretrain, feature UDA methods and joint semi-supervision.

Frequent classes achieve strong IoU even in the zero-shot setting; performance improvements are primarily observed for the *big_rock* class.

5.6 Per-class analysis

Feature-level UDA improves frequent-class IoU on the MER test, with method-dependent trade-offs between macro mIoU and weighted IoU. Table 5 breaks down CORAL, MMD, DANN and ADDA on MER test with per-class IoU, mIoU and wIoU. Comparing zero-shot pretraining with joint semi-supervision shows gains across all four terrain classes, with the largest relative improvement on *big_rock*. Table 6 compares zero-shot pretrain and joint semi-supervision on MER test. Improvements for the *big_rock* class remain limited in comparison to the frequent classes.

5.7 Discussion of results

Pseudo-labeling with an EMA teacher reaches moderate mIoU but remains inferior to joint semi-supervision. The two-phase semi-supervised approach lets pseudo-label errors to accumulate between labeled epochs. CORAL underperforms relative to MMD and DANN on MER mIoU, while ADDA attains the highest wIoU among UDA methods.

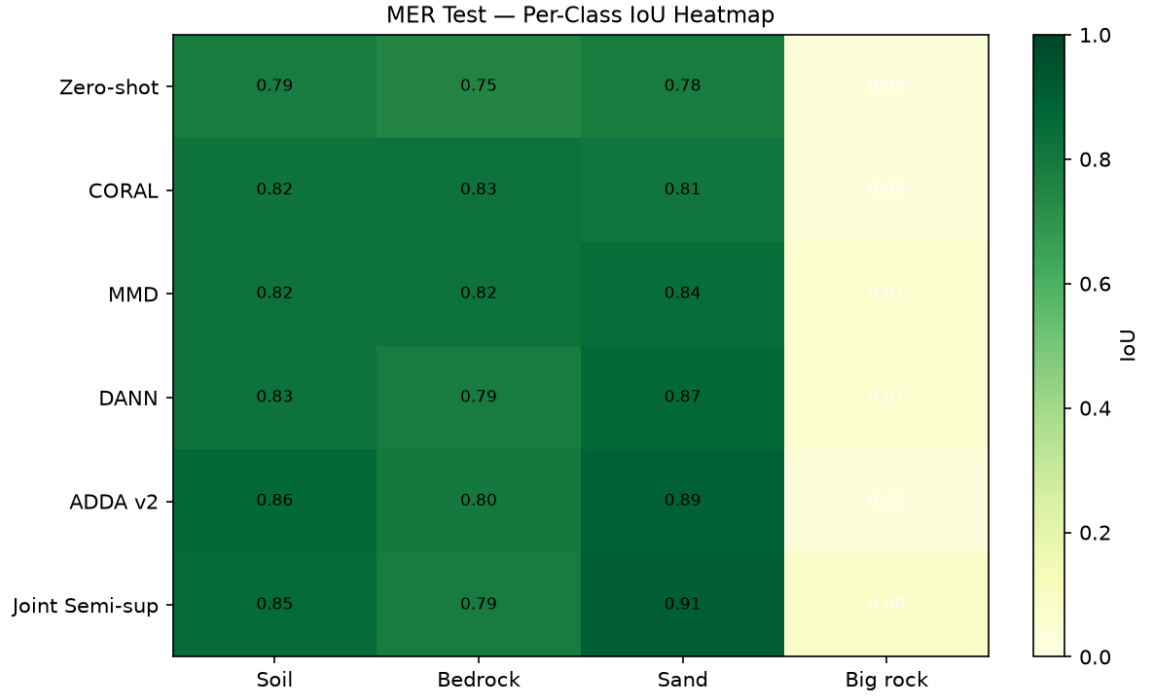


Figure 8: Per-class IoU heatmap on MER test (six key methods).

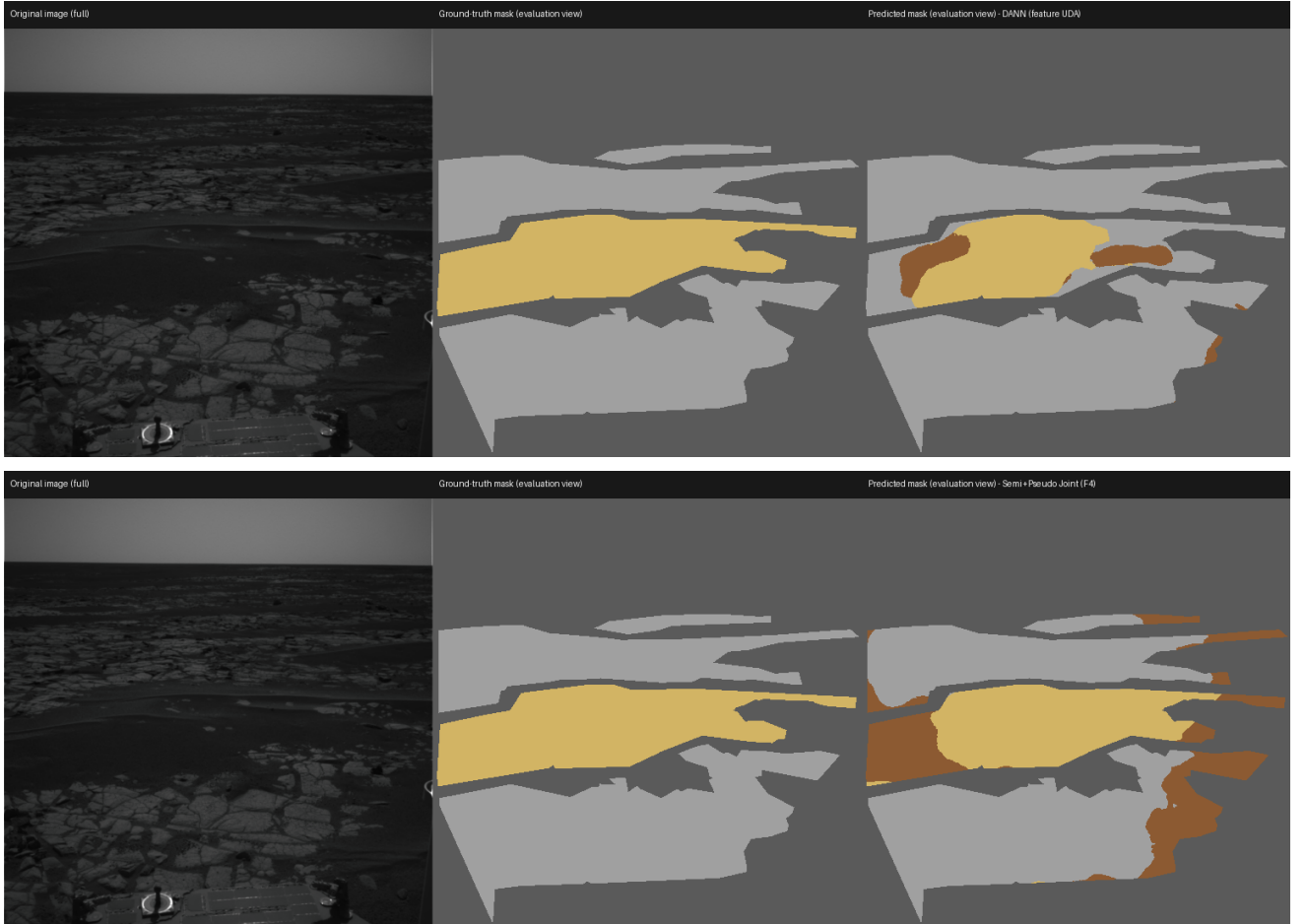


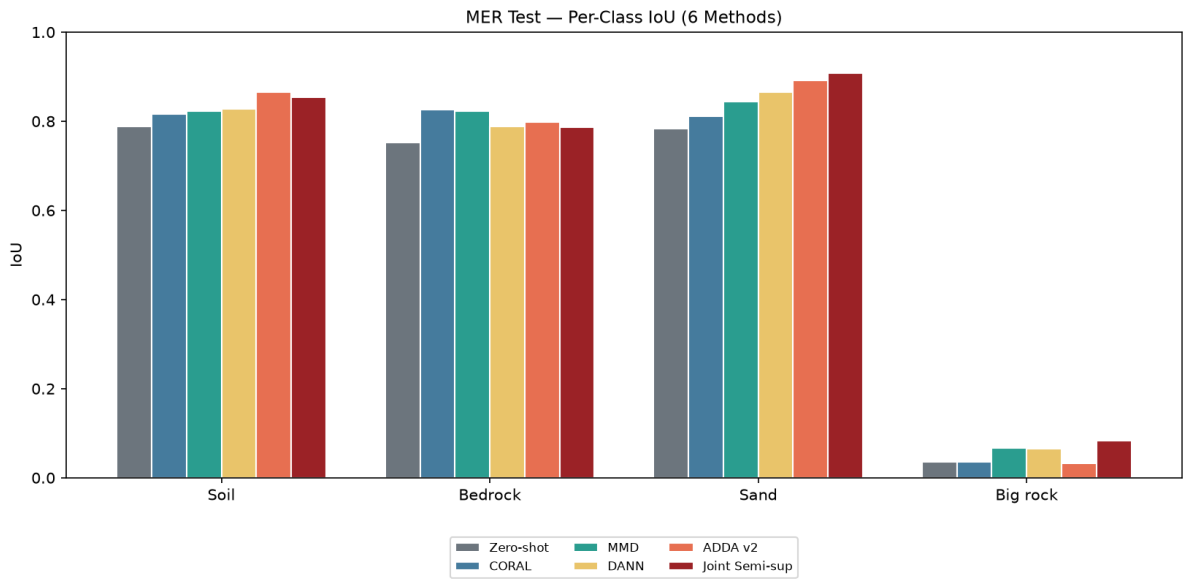
Figure 9: Qualitative MER test predictions after adaptation (image, GT and prediction). Top: DANN feature UDA. Bottom: joint semi-supervision.

Table 5: Feature-level UDA on MER test: per-class IoU, mIoU and wIoU.

Method	soil	bedrock	sand	big_rock	mIoU	wIoU
CORAL	0.817	0.827	0.812	0.036	0.623	0.810
MMD	0.823	0.823	0.844	0.066	0.639	0.821
DANN	0.828	0.789	0.866	0.066	0.637	0.820
ADDA	0.865	0.798	0.893	0.033	0.647	0.846

Table 6: Per-class IoU on MER test (zero-shot pretrain vs joint semi-supervision).

Class	Zero-shot (pretrain)	Joint semi-supervision
soil	0.788	0.855
bedrock	0.752	0.787
sand	0.784	0.907
big_rock	0.036	0.084
mIoU	0.590	0.658
wIoU	0.769	0.844

**Figure 10:** Per-class IoU on MER test across methods (grouped bars).

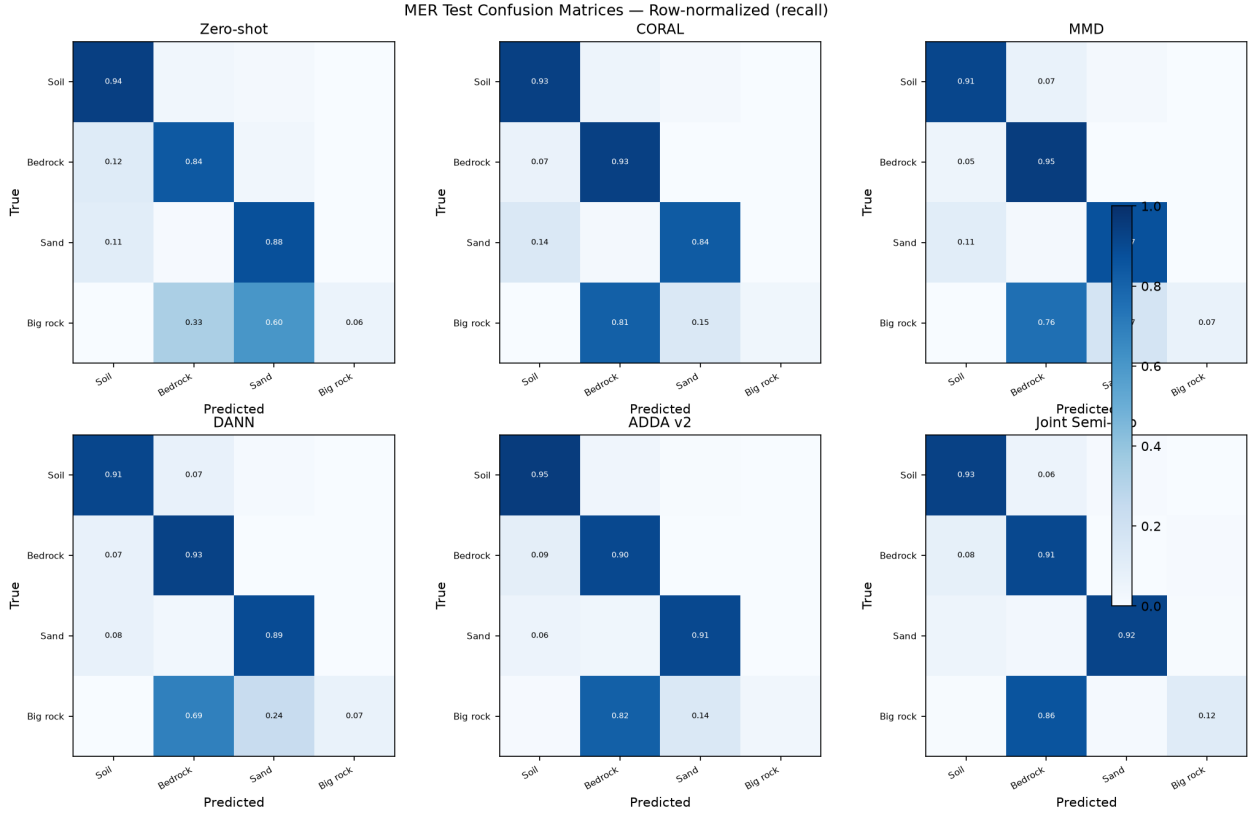


Figure 11: Row-normalized confusion matrices on MER test (class recall).

6 Conclusions

We addressed cross-rover Mars NavCam segmentation with a ResNet34-U-Net pretrained on MSL and adapted to MER using feature-level UDA and semi-supervised joint training on AI4MARS. Zero-shot transfer from pretraining establishes a usable MER baseline. Feature UDA improves MER test mIoU over zero-shot, with ADDA also yielding the highest UDA wIoU. Joint semi-supervision gives the largest gain by combining labeled MER images with unlabeled frames. The *big_rock* class remains difficult because it covers only a tiny fraction of pixels; mIoU gains partly reflect rare-class recovery under joint semi-supervision.

Joint semi-supervision leads on macro mIoU over zero-shot, while ADDA leads on wIoU; only joint semi-supervision substantially improves zero-shot *big_rock* IoU among the methods we consider deployable (zero-shot pretrain plus feature UDA and joint semi-supervision). Figure 12 consolidates these comparisons on MER test.

6.1 Future work

1. SAM or CLIP encoders could provide pseudo-masks or semantic targets for unlabeled MER frames, complementing the current EMA self-training loop without replacing the lightweight U-Net deployment path.
2. `m2020_val` (Perseverance rover) is not considered in our protocol; evaluating if MSL→MER models transfer to M2020 without retraining would test cross-mission robustness beyond the Spirit/Opportunity target.

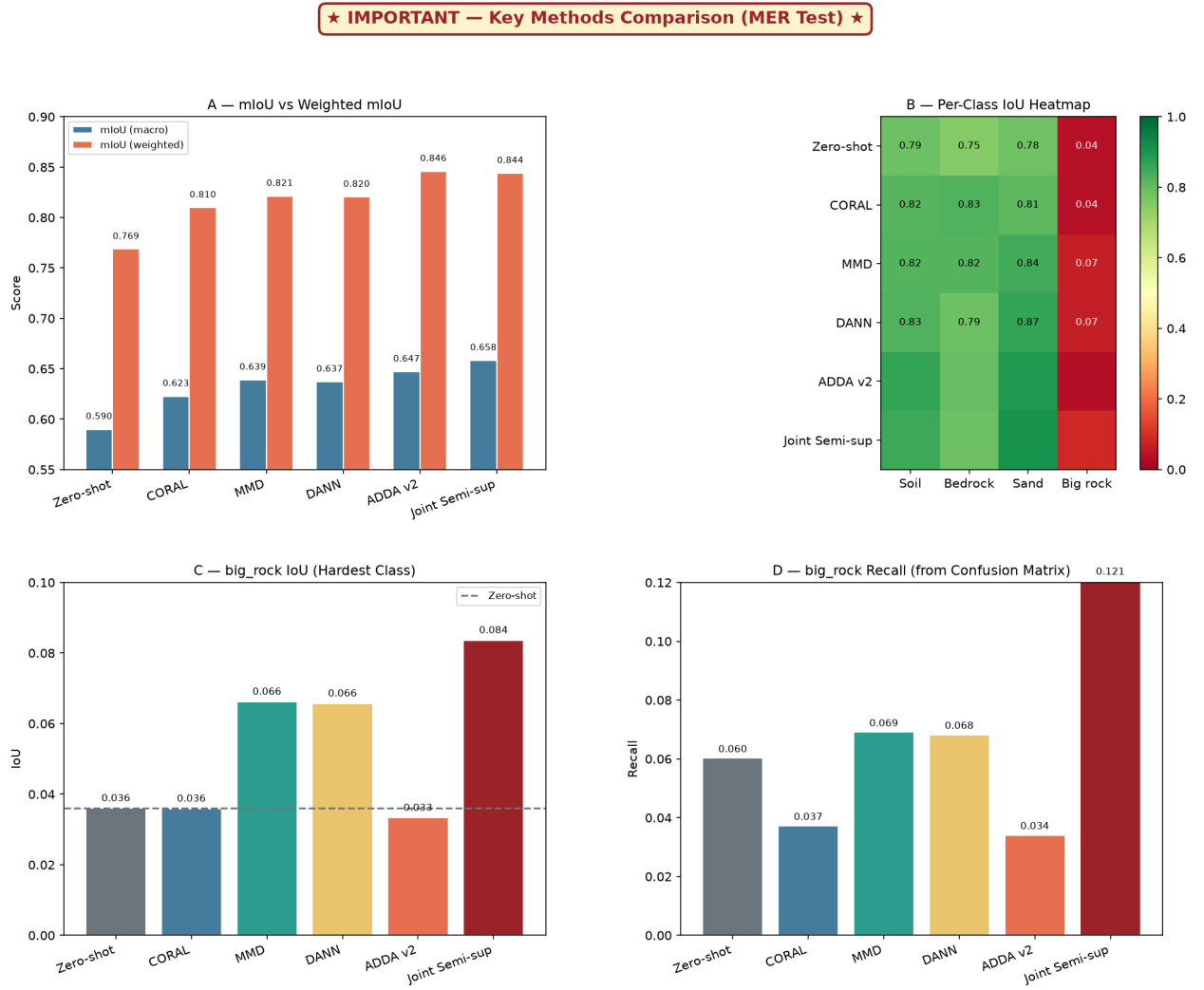


Figure 12: Summary comparison on MER test: mIoU vs. wIoU (A), per-class IoU heatmap (B), *big_rock* IoU (C) and *big_rock* recall from confusion matrices (D).

References

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